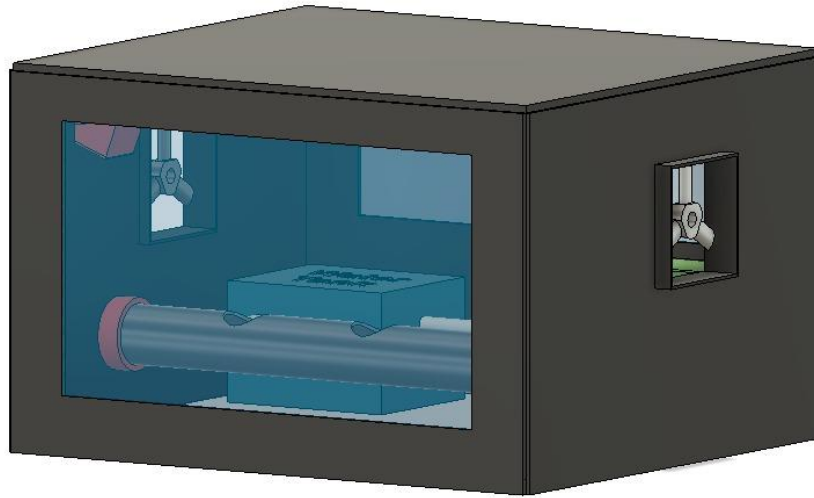


Smart Hydroponic System Using IOT

INTRODUCTION



Agriculture has been the backbone of human civilization for thousands of years, serving as the primary source of food production and livelihood for billions of people across the globe. Traditional soil-based agriculture has sustained humanity through different eras, but it comes with significant limitations that are becoming increasingly problematic in the modern age. Conventional farming is heavily dependent on several critical factors such as availability of arable land, quality of soil, adequate water supply, favorable climate conditions, and the absence of pests and diseases. With the global population projected to exceed 9 billion by 2050, the demand for food is growing exponentially, while the resources required for traditional farming are becoming scarce. Issues such as soil degradation, water scarcity, unpredictable weather patterns due to climate change, and shrinking agricultural land due to urbanization have

created an urgent need for alternative farming methods that are more efficient, sustainable, and less dependent on natural resources. Additionally, traditional farming methods often involve excessive use of chemical fertilizers and pesticides, which not only degrade soil health over time but also pose serious environmental and health risks. The inefficiencies of conventional agriculture become even more apparent in urban and semi-urban areas where space is limited, and soil quality is often poor or contaminated. In such scenarios, the ability to grow crops without soil and with minimal water usage becomes a revolutionary concept.

Hydroponics is an innovative agricultural technique that addresses many of the limitations associated with traditional soil-based farming. The term "hydroponics" is derived from the Greek words "hydro" meaning water and "ponos" meaning labor, essentially translating to "working water." In a hydroponic system, plants are grown without soil, and instead, their roots are immersed in or exposed to a nutrient-rich water solution that provides all the essential minerals and nutrients required for healthy growth. This method eliminates the dependency on soil quality and allows for precise control over the nutrients that plants receive, leading to faster growth rates, higher yields, and better quality produce. Hydroponic systems can be set up virtually anywhere—indoors, on rooftops, in greenhouses, or even in areas with poor soil conditions—making them highly versatile and adaptable to different environments. Because the roots have direct access to nutrients dissolved in water, plants do not need to expend energy searching for nutrients in the soil, which accelerates their growth cycle. Moreover, hydroponic farming uses up to 90% less water compared to traditional agriculture because the water in the system is recirculated and reused, with minimal loss due to evaporation or

runoff. This makes hydroponics particularly attractive in regions facing water scarcity or drought conditions. Additionally, since there is no soil involved, the risks associated with soil-borne diseases, pests, and weeds are significantly reduced, leading to healthier plants and reduced need for chemical pesticides.

However, despite its numerous advantages, hydroponic farming is not without challenges. One of the major obstacles is the need for continuous and precise monitoring of various environmental and nutrient parameters. Plants grown hydroponically are highly sensitive to changes in their growing environment, and even slight deviations in factors such as pH level, nutrient concentration, temperature, humidity, and light intensity can adversely affect plant health and productivity. For instance, if the pH level of the nutrient solution is too high or too low, it can hinder the plant's ability to absorb certain nutrients, leading to deficiencies and stunted growth. Similarly, improper nutrient concentration can result in either nutrient burn or malnutrition. Temperature and humidity also play crucial roles in photosynthesis and transpiration processes, and maintaining them within optimal ranges is essential for healthy plant development.

In summary, the proposed Smart Hydroponic System using IoT combines the proven benefits of soil-less agriculture with the power of modern digital technology to create a farming solution that is efficient, sustainable, scalable, and user-friendly. It represents a practical step toward addressing the challenges of modern agriculture and paving the way for a future where food production is less dependent on natural resources and more resilient to environmental uncertainties.

OBJECTIVES

The primary objective of this project is to develop a Smart Hydroponic System that leverages Internet of Things (IoT) technology to automate and optimize the cultivation of plants in a soil-less environment. The focus is on creating a system that can continuously monitor critical environmental parameters such as pH level, temperature, humidity, and light intensity without requiring constant human supervision. By enabling real-time data acquisition and automated control, the system aims to maintain optimal growing conditions for plants at all times, ensuring faster growth rates, higher yields, and better quality produce. The intention is to eliminate the inefficiencies and human errors associated with manual monitoring and adjustment, thereby making hydroponic farming more accessible, reliable, and productive for both hobbyists and commercial growers. This automation is particularly crucial in hydroponics where even slight deviations in environmental conditions can significantly impact plant health and productivity.

Another major objective is the integration of multiple sensors to enable comprehensive environmental monitoring that captures all aspects of the growing environment. Instead of relying on periodic manual checks or single-parameter monitoring, the system will be equipped with a pH sensor to measure nutrient solution acidity, a DHT22 sensor to track temperature and humidity levels, and a light intensity sensor to monitor illumination conditions. This multi-sensor approach ensures that all critical factors affecting plant growth are continuously tracked with precision and accuracy. By doing so, the system can identify problems early, prevent nutrient deficiencies, avoid

environmental stress, and maintain the delicate balance required for healthy plant development. The sensor integration is critical to making the system truly intelligent and capable of responding to the complex and dynamic conditions present in a hydroponic setup where variables are constantly changing based on plant growth stage, time of day, and external environmental factors.

A further objective is to implement cloud-based data logging and remote accessibility through a web-based interface that allows users to monitor and control their hydroponic system from anywhere in the world. Through IoT connectivity, the system can transmit sensor data to a cloud platform where it is stored, processed, and made available for visualization and analysis. Users can access real-time information about their plants' growing conditions using a smartphone, tablet, or computer, regardless of their physical location. Parameters such as current pH level, temperature, humidity, light intensity, and system status can be monitored remotely, ensuring efficient management and timely intervention when needed. The web interface also enables users to view historical data in the form of graphs and charts, track trends over time, configure system settings, and receive alerts when parameters exceed safe thresholds. This combination of automation and remote monitoring provides a balance between independence and oversight, ensuring that users remain informed and in control even when they are away from their growing location.

LITERATURE SURVEY

1) Proença, P. F., & Simões, P. (2020) — TACO: Trash Annotations in Context

What they did: Released a comprehensive real-world litter dataset (TACO) containing over 1,500 images with instance segmentation labels covering 60 categories of waste items. They benchmarked several deep learning models including Mask R-CNN for trash detection and emphasized the importance of diverse backgrounds, lighting conditions, and real-world scenarios

What we took: The concept of using diverse, real-world annotated datasets to train AI models for environmental monitoring inspired our approach to integrating vision-based classification in the hydroponic system. While our focus is on plant health monitoring rather than trash detection, we adopted the principle of using pre-trained models and transfer learning with domain-specific datasets.

2) Mittal, G., Yagnik, K. B., Garg, M., & Krishnan, N. C. (2016) — SpotGarbage: Smartphone App to Detect Garbage Using Deep Learning

What they did: Developed a smartphone-based application that uses convolutional neural networks (CNNs) to detect and classify garbage in real-time using the phone's camera. They collected a custom dataset of garbage images and trained models to run efficiently on mobile devices with limited computational resources.

What we took: The idea of deploying AI-based image recognition on resource-constrained embedded devices directly influenced our decision to use the ESP32-CAM module with cloud-based AI processing via the Gemini API. We adopted their approach of offloading heavy computational tasks to more powerful processors

3) Saleh, M. A., Pothalaiah, S., & Veeramanickam, M. (2021) — IoT-Based Monitoring System for Hydroponic Vertical Farming

What they did: Designed and implemented an IoT-enabled monitoring system for vertical hydroponic farms that measured parameters such as water level, pH, electrical conductivity (EC), temperature, and humidity using Arduino and NodeMCU microcontrollers.

What we took: This study provided valuable insights into sensor selection, calibration procedures, and cloud platform integration for hydroponic applications. We adopted their methodology of using ThingSpeak or similar IoT platforms for data logging and visualization. Their approach to implementing threshold-based alerts influenced our decision to incorporate notification systems that inform users when environmental conditions deviate from optimal ranges.

4) Saaid, M. F., Yahya, N. H. N., Noor, M. Z. H., & Ali, M. S. A. M. (2015) — A Development of an Automatic Microcontroller System for Deep Water Culture (DWC)

What they did: Developed an automated Deep Water Culture (DWC) hydroponic system controlled by a PIC microcontroller. The system monitored pH and electrical conductivity using industrial-grade sensors and implemented automated nutrient dosing using peristaltic pumps.

What we took: Their automated pH and EC control mechanism provided a blueprint for our automated control subsystem. We adopted the concept of using relay-controlled peristaltic pumps for precise liquid dosing and implemented similar threshold-based control logic for activating corrective actions.

COMPONENTS USED :

Component	Description	Purpose
ESP32 Microcontroller	Wi-Fi and Bluetooth-enabled microcontroller with multiple GPIO pins	Acts as the brain of the system, processes sensor data, and enables IoT connectivity
pH Sensor Module	Analog pH sensor with probe and signal conditioning circuit	Measures the acidity or alkalinity of nutrient solution
DHT11 Sensor	Digital temperature and humidity sensor	Monitors air temperature and relative humidity in the growing environment
LDR Sensor	Light Dependent Resistor sensor	Measures ambient or artificial light levels to ensure adequate illumination
Relay Module (4-Channel)	Electrically operated switches for controlling AC/DC devices	Enables microcontroller to control high-power devices like pumps, fans, and lights
Water Pump	Submersible or inline pump for liquid circulation	Circulates nutrient solution through the hydroponic system
12V DC Power Supply	Regulated power adapter	Powers the ESP32, sensors, relays, and other electronic components
Cooling Fan	12V DC axial fan	Regulates temperature and improves air circulation in the growing area
LED Grow Lights	Full-spectrum LED lighting system	Provides artificial illumination for plant growth when natural sunlight is insufficient
Connecting Wires	Jumper wires and electrical cables	Establishes electrical connections between components
PCB (Printed Circuit Board)	Custom printed circuit board	Provides a platform for assembling and organizing electronic components
Enclosure Box	Waterproof plastic housing	Protects electronic components from moisture, dust, and physical damage
PVC Pipes/Grow Channels	Hollow pipes or channels	Holds plants and allows nutrient solution to flow to plant roots

1. ESP32 MICROCONTROLLER



Overview:

- Low-cost, powerful microcontroller by Espressif Systems
- Ideal for IoT applications with built-in connectivity

Key Features:

- Dual-core 32-bit processor (up to 240 MHz)
- Built-in Wi-Fi (802.11 b/g/n) and Bluetooth
- Up to 34 GPIO pins for sensors and actuators
- Multiple communication protocols: SPI, I2C, UART
- 520 KB SRAM, supports external flash memory
- Multiple ADC channels for analog sensor.

2. pH SENSOR MODULE



Overview:

- Measures acidity/alkalinity of nutrient solution
- pH scale: 0 (acidic) to 14 (alkaline), 7 is neutral

Working Principle:

- Glass electrode generates voltage based on pH
- Reference electrode provides stable potential
- Signal conditioning circuit converts to readable voltage

Specifications:

- Measurement range: pH 0-14
- Accuracy: ± 0.1 pH units
- Resolution: 0.01 pH
- Response time: 1-2 seconds.

3. DHT11 TEMPERATURE AND HUMIDITY SENSOR



Overview:

- Digital sensor for temperature and humidity
- Also known as AM2302

Specifications:

- Temperature range: -40°C to $+80^{\circ}\text{C}$
- Temperature accuracy: $\pm 0.5^{\circ}\text{C}$
- Humidity range: 0-100% RH
- Humidity accuracy: $\pm 2-5\%$ RH
- Sampling rate: Once every 2 seconds

Application:

- Monitors air temperature and humidity
- Optimal temperature: $20-26^{\circ}\text{C}$
- Optimal humidity: 60-70% RH
- Triggers fans or humidifiers
- Critical for plant health

4. LDR (Light Dependent Resistor):



Overview:

- Resistance decreases with light intensity
- Simple and low-cost
- Requires voltage divider circuit
- Non-linear response
- Good for basic light detection

Application:

- Monitors natural and artificial light
- Controls grow lights automatically
- Ensures adequate light for photosynthesis
- Leafy greens need: 10,000-20,000 lux
- Fruiting plants need: 30,000-50,000 lux
- Implements day/night cycles (12-18 hours).

5. RELAY MODULE (4-CHANNEL)



Overview:

- Electrically operated switches
- Allows low-voltage control of high-power devices

Working Principle:

- Electromagnetic coil operates mechanical switch
- Provides electrical isolation
- NO (Normally Open), NC (Normally Closed), COM terminals

Application:

- Controls water pump on/off
- Switches cooling fan
- Controls grow lights
- Each relay handles one device independently

6. 12V DC POWER SUPPLY



Overview:

- Converts AC mains (110V/220V) to 12V DC
- Powers all electronic components

Types:

- **Linear:** Clean output, less efficient, larger
- **SMPS (Switched-Mode):** Efficient, compact, common

Specifications:

- Output voltage: 12V DC regulated
- Current rating: Must exceed total system draw
- Protection: Overcurrent, overvoltage, short-circuit

Application:

- Powers all system components reliably
- Stable power for accurate sensor readings
- Continuous operation capability

8. COOLING FAN



Overview:

- Regulates temperature in growing environment
- Typically 12V DC axial fans

Features:

- Axial design for high airflow
- Ball bearings for longer life
- Variable speed control (PWM possible)

Application:

- Activated when temperature exceeds threshold
- Controlled via relay
- Maintains optimal temperature (20-26°C)
- Improves air circulation and CO₂ exchange
- Reduces humidity to prevent disease

9. LED GROW LIGHTS



Overview:

- Artificial lighting for plant growth
- Energy-efficient light source

Advantages:

- Energy efficient
- Long lifespan (50,000+ hours)
- Low heat output
- Customizable spectrum

Application:

- Provides consistent light indoors
- Controlled via relay
- Timer or sensor-based operation
- Enables year-round growing

10. 12V Water Pump



Description:

- Operates on 12V DC power supply
- Can be submersible (placed inside water) or inline (external to water)
- Compact and energy-efficient design
- Typically rated between 3-6 liters per minute flow rate
- Low noise operation suitable for indoor systems

Purpose:

- Circulates nutrient solution throughout the hydroponic system
- Delivers water and nutrients directly to plant roots
- Maintains consistent flow of nutrient-rich water
- Prevents stagnation and ensures oxygen distribution
- Keeps nutrients evenly mixed in the solution

Key Features:

- Waterproof and corrosion-resistant construction
- Compatible with standard tubing sizes
- Low power consumption (typically 4-10 watts)

Software Tools Used:

Software Tool	Description	Purpose
Arduino IDE	Integrated Development Environment for ESP32-CAM programming	Write and upload embedded code, control camera, motors, and sensors, debug using serial monitor
HTTP Protocol	Standard web communication protocol for client-server data exchange	Send captured images to server, communicate with Gemini API, upload sensor data, receive control commands
Camera (ESP32-CAM)	OV2640 camera module for image capture and video streaming (up to 2MP)	Capture real-time images of surroundings, detect trash objects, stream to AI for analysis, visual navigation feedback
WebSocket Protocol	Full-duplex real-time bidirectional communication protocol	Real-time video streaming, live status updates, instant command delivery, continuous sensor data streaming with low latency
Custom Server	Backend server (Node.js/ Python/Java) for request handling and coordination	Receive images from ESP32, forward to Gemini AI, process responses, send control commands, host web dashboard
Gemini AI API	Google's advanced AI for image recognition and object detection	Analyze captured images, identify trash objects, classify trash types, distinguish trash from non-trash, intelligent decision making
Fusion 360	Professional 3D CAD/CAM software for design and modeling	Design robot chassis, model robotic arm, create mounting brackets, design trash container, export STL for 3D printing
Visual Studio Code	Free, powerful code editor with extensive extension support	Write ESP32 firmware, develop web dashboard, create server code, debug and test, version control, project management

1. ARDUINO IDE

Overview:

- Integrated Development Environment for programming microcontrollers
- Open-source software for ESP32 programming
- User-friendly interface for beginners and professionals

Key Features:

- Built-in code editor with syntax highlighting
- Serial monitor for real-time debugging
- Library manager for easy dependency installation

- Board manager for ESP32 support
- One-click compilation and upload
- Cross-platform compatibility (Windows, Mac, Linux)

Application:

- Write embedded code for ESP32-CAM
- Configure camera settings and capture images
- Implement obstacle detection logic
- Setup Wi-Fi connectivity
- Control servo motors and DC motors
- Upload firmware to microcontroller
- Debug using serial monitor

2. HTTP PROTOCOL

Overview:

- HyperText Transfer Protocol for web communication
- Standard protocol for client-server communication
- Request-response model for data exchange

HTTP Methods:

- GET: Retrieve data from server
- POST: Send data to server
- PUT: Update existing data
- DELETE: Remove data from server

Request/Response Structure:

- URL/endpoint specification
- HTTP headers (Content-Type, Authorization)
- Status codes indicate success or failure
- Data typically in JSON or XML format

3. CAMERA (ESP32-CAM)

Overview:

- Integrated camera module on ESP32-CAM board
- OV2640 camera sensor (2 megapixels)
- Captures images and streams video

Camera Configuration:

- Resolution settings (QVGA, VGA, SVGA, UXGA)
- Brightness and contrast adjustment
- White balance settings
- Exposure control
- Image quality settings (JPEG compression)

Application:

- Continuously capture images of surroundings
- Detect trash objects in real-time
- Stream images to Gemini AI for analysis
- Visual feedback for navigation

4. WEBSOCKET PROTOCOL

Overview:

- Full-duplex communication protocol over TCP
- Enables real-time bidirectional data transfer
- Persistent connection between client and server

Key Features:

- Low latency communication
- Reduced overhead compared to HTTP polling
- Real-time data streaming
- Event-driven architecture
- Single TCP connection for both directions

Application:

- Real-time video streaming from ESP32-CAM
- Live status updates to dashboard
- Instant command delivery to robot
- Continuous sensor data streaming
- Real-time trash detection notifications
- Remote control with minimal latency
- Live monitoring of robot position and status

5. CUSTOM SERVER

Overview:

- Backend server for handling requests and data processing
- Intermediary between ESP32-CAM and cloud services
- Custom logic for system coordination

Server Technologies:

- Node.js with Express framework
- Python with Flask/Django
- Java with Spring Boot
- Choice based on requirements and expertise

Application:

- Central hub for system communication
- Coordinate between ESP32 and Gemini AI
- Manage data flow and storage
- Host web dashboard interface
- Implement business logic
- Monitor system health
- Generate reports and analytics

6. GEMINI AI API (FOR IMAGE PROCESSING)

Overview:

- Google's advanced AI model for image recognition
- Cloud-based service for object detection and classification

- Powerful computer vision capabilities

API Features:

- RESTful API interface
- JSON request and response format
- Base64 encoded image upload
- Batch processing support
- Configurable confidence thresholds

Advantages:

- High accuracy in object recognition
- Pre-trained on massive datasets
- No local GPU required
- Continuous model improvements
- Multi-language support

Application:

- Real-time trash identification
- Distinguish trash from non-trash objects
- Classify trash categories
- Reduce false positives
- Intelligent decision making
- Adaptive learning from results
- Quality assurance for collection

7. FUSION 360

Overview:

- Professional 3D CAD/CAM software by Autodesk
- Cloud-based design and modeling platform
- Free for students and hobbyists

Key Features:

- Parametric 3D modeling
- Assembly design and constraints
- Simulation and analysis tools.

Design Capabilities:

- Sketch-based 3D modeling
- Extrude, revolve, sweep, loft operations
- Boolean operations (union, subtract, intersect)
- Fillet and chamfer tools
- Pattern and mirror features
- Sheet metal design

8. VISUAL STUDIO CODE (VS CODE)

Overview:

- Free, open-source code editor by Microsoft
- Lightweight yet powerful development environment
- Extensive extension ecosystem

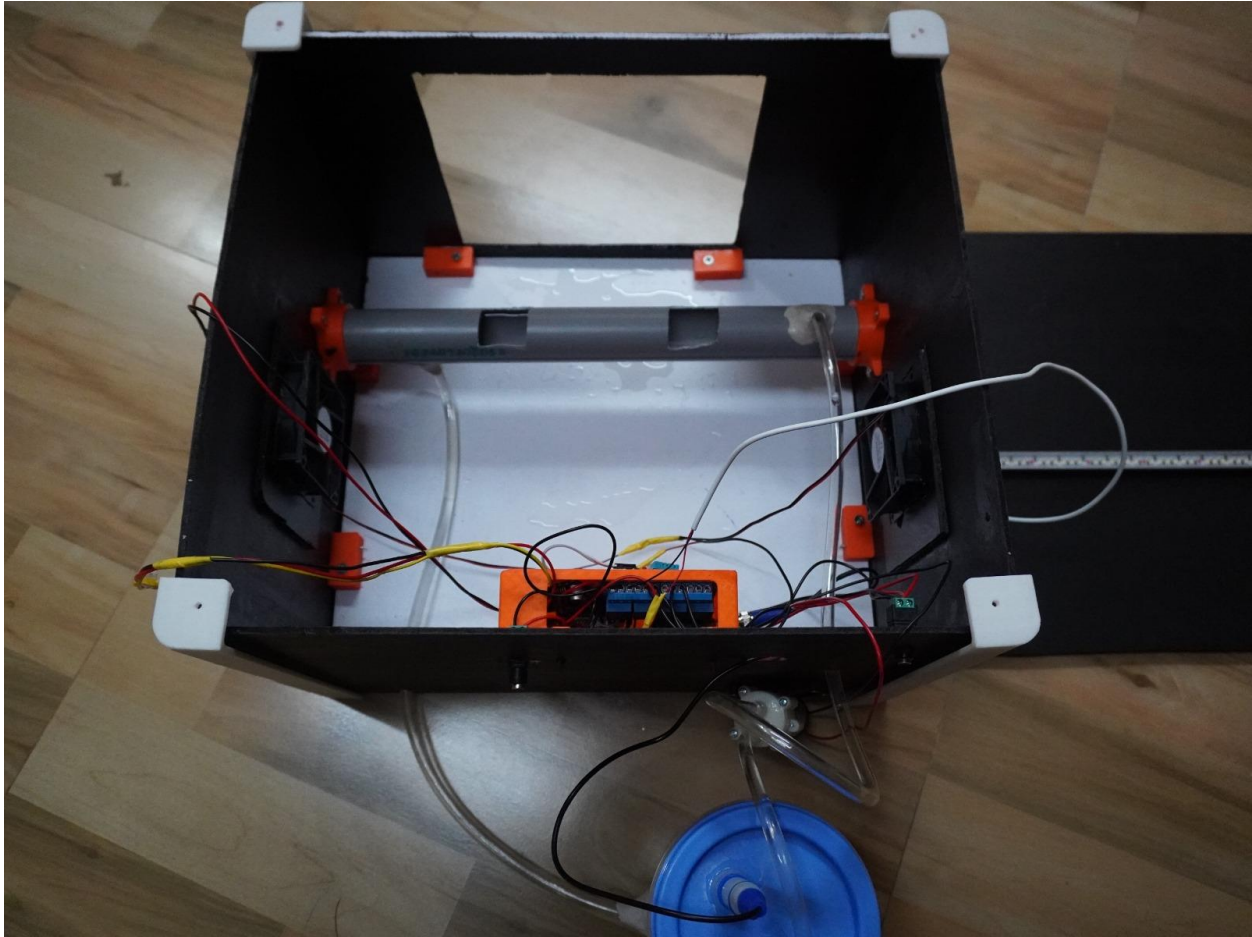
Key Features:

- IntelliSense (intelligent code completion)
- Syntax highlighting for multiple languages
- Integrated Git version control
- Built-in terminal
- Debugging tools
- Split view editing
- Live server extension for web development

Application:

- Write and edit ESP32 firmware code
- Develop web dashboard (HTML/CSS/JS)
- Create custom server code
- Edit configuration files
- Debug and test code
- Manage project files
- Version control with Git
- Documentation writing
- API testing and development

WORKING PRINCIPLE



The Smart Hydroponic System using IoT operates on the fundamental principle of continuous environmental monitoring, automated control, cloud-based data management, and remote accessibility to create an optimized growing environment for plants without soil. The entire operation can be understood as a closed-loop feedback system where sensors continuously measure critical parameters, the microcontroller processes this data and makes control decisions, actuators adjust environmental conditions, and all information is transmitted to the cloud for remote monitoring and historical analysis.

SYSTEM INITIALIZATION AND STARTUP

The system begins its operational cycle when power is supplied from the 12V DC power supply. The ESP32 microcontroller initiates its boot sequence, loading the embedded firmware from flash memory and systematically initializing all connected sensors and control devices. The microcontroller configures its GPIO pins according to their designated functions, setting appropriate pins as digital inputs for sensor data, digital outputs for relay control, and analog inputs for sensors like the pH probe.

Once hardware initialization completes, the ESP32 establishes network connectivity by scanning for the pre-configured Wi-Fi network. The microcontroller transmits authentication credentials and completes the handshake protocol with the wireless access point. Upon successful connection, the system either receives an IP address from DHCP or configures itself with a static IP. The ESP32 then connects to the cloud platform, whether ThingSpeak, Firebase, or a custom server, by authenticating with API keys or service credentials. Once cloud connectivity is confirmed through a successful test transmission, the system enters normal operational mode and begins the continuous monitoring and control cycle.

DATA PROCESSING AND THRESHOLD EVALUATION

After acquiring sensor readings, the ESP32 processes this data to determine the current state of the growing environment and whether control actions are necessary. Each measured parameter is compared against predefined optimal ranges stored in firmware or configurable through the web interface. For most hydroponic crops, optimal pH ranges from 5.5 to 6.5, temperature from 20°C to 26°C, humidity from 60% to 70%, and light intensity depends on the specific crop requirements.

The threshold evaluation uses conditional logic to classify each parameter as acceptable, too high, or too low. For pH measurement, if the reading is below 5.5, the solution is too acidic and requires pH-up solution. If above 6.5, it's too alkaline and needs pH-down solution. Temperature evaluation determines if

cooling or heating is needed. If temperature exceeds 26°C, the cooling fan should activate. The system implements hysteresis to prevent rapid on-off cycling, meaning once the fan activates at 26°C, it remains on until temperature drops to perhaps 24°C.

AUTOMATED CONTROL AND ACTUATION

When data processing determines that control action is necessary, the ESP32 activates the appropriate actuator through the relay module. The relay acts as an electrically controlled switch allowing the low-voltage microcontroller to control high-power devices. When a control pin is set high, current flows through the relay coil, creating a magnetic field that closes the contacts and connects power to the controlled device. When set low, the coil de-energizes and contacts open, disconnecting power.

The water pump circulates nutrient solution through the hydroponic system. In NFT systems, the pump typically runs continuously. In ebb and flow systems, it operates on timed cycles. The ESP32 controls the pump relay based on system type and programmed schedules. The cooling fan relay activates when temperature exceeds the upper threshold. The fan operates until temperature drops below the lower hysteresis point, then deactivates. This on-off control effectively maintains temperature within acceptable bounds.

The grow light relay manages artificial illumination based on light sensor readings and time schedules. If natural light is insufficient during the photoperiod, grow lights activate. The ESP32 maintains time through internal clocks synchronized via internet NTP or external RTC modules. By comparing current time against programmed schedules, the system determines when to control lighting. If automated pH control is implemented, dosing pumps briefly activate to inject pH adjustment solutions when needed, with the system waiting for mixing and stabilization before re-measuring.

WEB-BASED MONITORING AND CONTROL INTERFACE

The web dashboard provides an intuitive interface for monitoring system status and controlling operations from any browser-enabled device. Implemented using HTML, CSS, and JavaScript, the dashboard retrieves data from the cloud platform using API calls and updates display elements to reflect current conditions. For Firebase systems, the JavaScript SDK enables real-time data binding where database changes automatically update the interface.

The dashboard includes numeric displays showing current sensor values with color coding indicating whether parameters are within optimal ranges. Gauge widgets provide visual representations using scales that show current values within expected ranges. Status indicators display on/off states of controlled devices. Historical data visualization through line charts plots sensor values against time, revealing patterns like daily temperature variations or gradual pH drift. Users can select different time ranges to focus on specific periods.

Control elements allow manual override of automated systems or remote configuration adjustments. Toggle switches enable users to manually control devices. Configuration inputs allow adjustment of optimal ranges, thresholds, and schedules to accommodate different plants or growth stages. Changes transmit to the cloud database, and the ESP32 periodically retrieves updated settings to implement new parameters. Responsive design ensures the dashboard functions effectively on all devices from desktops to smartphones.

CONTINUOUS FEEDBACK LOOP AND SYSTEM OPTIMIZATION

The Smart Hydroponic System operates as a continuous closed-loop control system where sensor inputs drive control decisions, control actions modify the environment, the modified environment affects sensor readings, and the cycle repeats indefinitely. This feedback loop enables automatic maintenance of optimal conditions despite external disturbances like ambient temperature changes, humidity variations, or pH drift. The system responds dynamically, activating cooling when heat builds up, adjusting lighting as daylight varies, and maintaining pH balance as plants absorb nutrients.

Machine learning techniques can potentially enhance performance by learning optimal control strategies from operational data. Predictive models could forecast temperature increases and enable preemptive cooling activation. Pattern recognition might identify optimal timing for nutrient changes. Anomaly detection could alert users to unusual readings indicating equipment failures. While these advanced capabilities require significant computational resources, cloud-based processing of uploaded data can implement analyses and provide recommendations or automatically adjust parameters through bidirectional IoT connection, enabling continuous system improvement and optimization for maximum plant growth and resource efficiency.

METHODOLOGY

The development and implementation of the Smart Hydroponic System using IoT follows a systematic and structured approach that encompasses requirement analysis, system design, component selection, hardware assembly, software development, sensor calibration, system integration, testing, deployment, and performance evaluation.

REQUIREMENT ANALYSIS AND SYSTEM DESIGN

The initial phase begins with identifying critical environmental parameters for optimal plant growth including pH level, air temperature, relative humidity, and light intensity. Research establishes optimal ranges for common hydroponic crops and determines necessary control actions such as nutrient circulation, temperature regulation, humidity management, and lighting control. A comprehensive system architecture is designed specifying the ESP32 as the central controller coordinating sensor inputs and control outputs. Block diagrams visualize component relationships showing power distribution, data flow from sensors through the microcontroller to the cloud, and control signals from the microcontroller through relays to actuators. Circuit diagrams specify exact electrical connections between components including pin assignments, voltage levels, and power distribution from the 12V supply through regulators to provide appropriate voltages for all components.

COMPONENT SELECTION AND PROCUREMENT

Components are selected based on technical requirements, cost, availability, compatibility, and reliability. The ESP32 microcontroller is chosen for integrated Wi-Fi, sufficient GPIO pins, ADC capability, and Arduino IDE compatibility. The pH sensor module is selected for appropriate measurement range, accuracy, and analog output compatibility. The DHT11 is chosen for digital output, reasonable accuracy, and simple communication protocol. The LDR is selected based on desired light measurement accuracy. The 4-channel relay module is selected to control all actuators with appropriate voltage and current ratings.

PHASE 3: HARDWARE ASSEMBLY AND CIRCUIT CONSTRUCTION

Hardware assembly begins with breadboard prototyping allowing quick modifications. The ESP32 is placed centrally with power connections established first. The 12V supply connects through voltage regulators to provide 5V and 3.3V rails. Common ground connections are established for all components. The pH sensor analog output connects to an ESP32 ADC pin. The DHT22 connects to a digital GPIO pin with a pull-up resistor. Relay module control inputs connect to digital GPIO pins, with the relay module powered separately from the 12V supply. High-power actuators connect through relay contacts. Wire management ensures organized, reliable connections. All connections are verified against circuit diagrams using multimeter continuity testing before applying power. Once validated, the circuit can be transferred to a permanent PCB with soldered connections for improved reliability.

SENSOR CALIBRATION

Sensor calibration ensures measurement accuracy essential for effective control. The pH sensor requires careful calibration using standard buffer solutions with known pH values. The probe is rinsed, immersed in pH 7.0 buffer, and the ADC value is recorded. The process repeats with pH 4.0 buffer to obtain a second reference point. Using these two points, a linear calibration equation is derived calculating the slope and offset. These calibration coefficients are stored in firmware. The calibration is verified by testing in solutions of known pH. The DHT11 comes pre-calibrated but can be verified against reference instruments with software offset corrections applied if needed. If using an LDR, calibration involves measuring voltage at several known light intensities and creating a lookup table or calibration curve.

SOFTWARE DEVELOPMENT AND PROGRAMMING

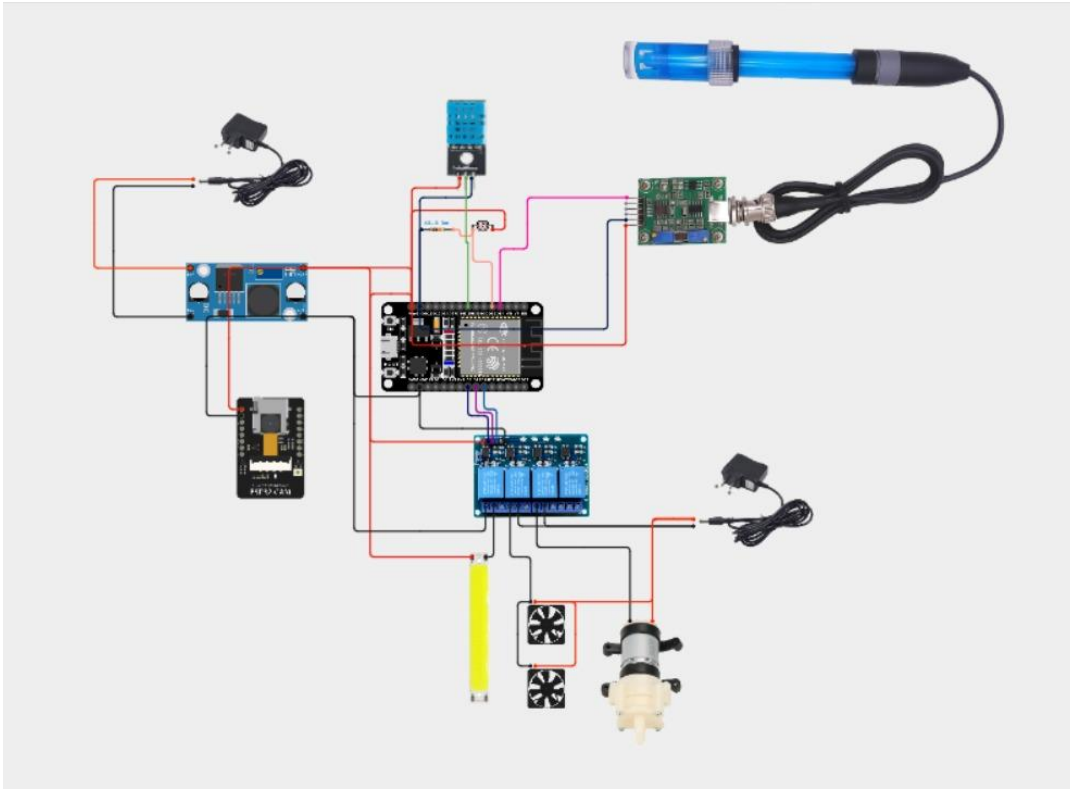
Firmware is developed using Arduino IDE with ESP32 board support installed. The code includes necessary libraries for WiFi, sensors, and cloud platforms. Constants define pin assignments, WiFi credentials, and cloud API keys. The setup function initializes serial communication, configures GPIO pins, establishes WiFi connection, initializes sensors, and sets relay default states.

The main loop reads all sensors by retrieving ADC values for pH and processing through calibration equations, calling DHT library functions for temperature and humidity, and reading light intensity. Threshold evaluation logic compares readings against stored limits using conditional statements. Control logic activates or deactivates relays through digitalWrite commands based on threshold comparisons, implementing hysteresis to prevent rapid switching. Data transmission formats sensor readings into appropriate structures for the cloud platform, constructing HTTP requests for ThingSpeak or JSON objects for Firebase. Error handling manages WiFi disconnections, failed sensor readings, and system hangs. The code includes appropriate delays and comments for clarity and maintainability.

SYSTEM INTEGRATION AND TESTING

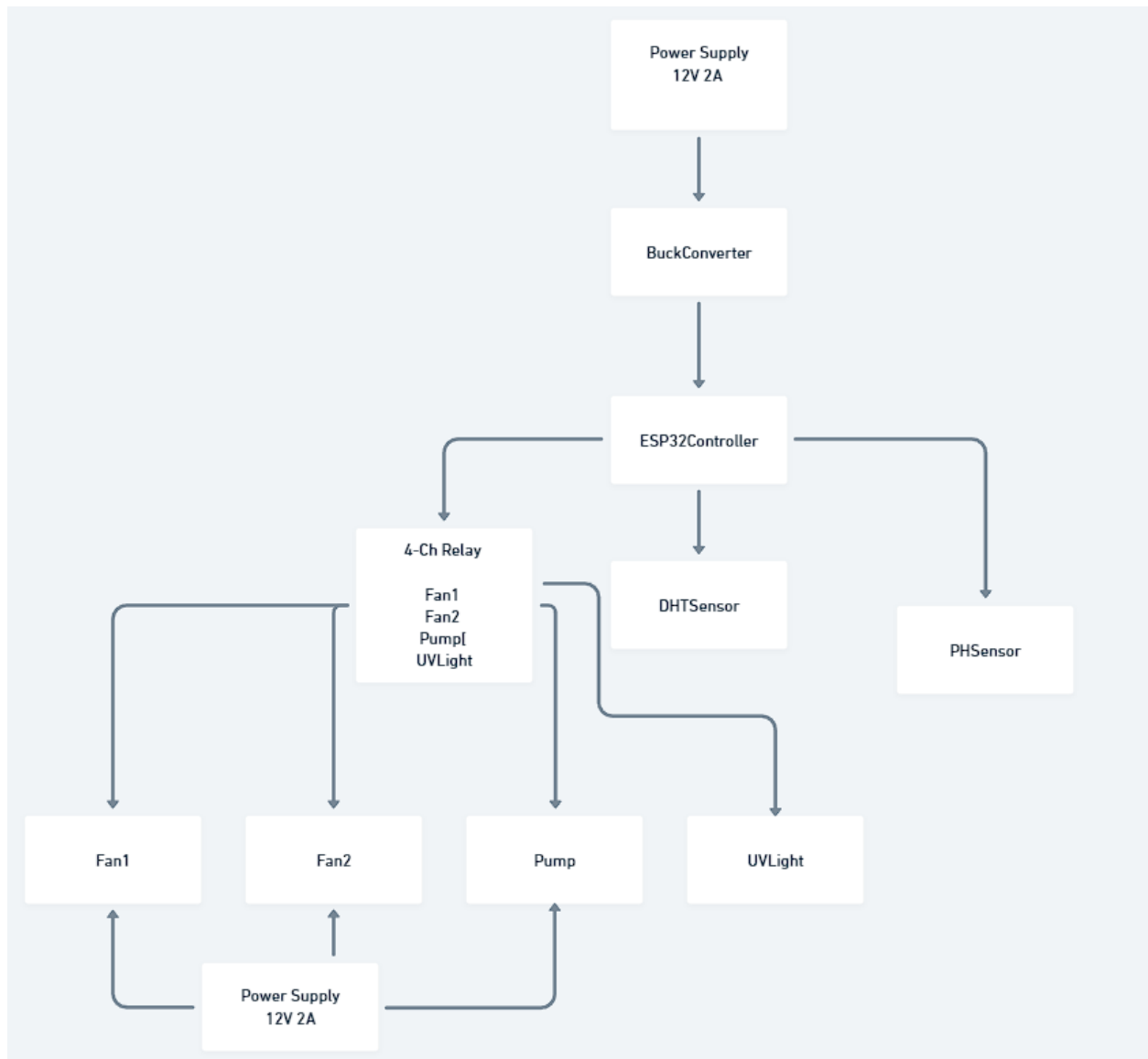
Hardware is finalized with all components mounted in appropriate locations. Sensors are positioned properly: pH probe in reservoir, DHT11 at canopy height, light sensor measuring plant-level illumination. Actuators are installed with correct positioning and connections. Individual component testing verifies each sensor reports correct values when exposed to known conditions and each actuator operates when manually activated. Integration testing runs complete operational cycles, artificially creating conditions to verify automatic responses such as heating to trigger cooling fan or adjusting pH to test dosing. Network communication testing confirms data transmits to cloud and appears correctly in databases and dashboards accessed from multiple devices. Stress testing evaluates behavior during network disruptions, power interruptions, and sensor failures. Extended continuous operation identifies long-term reliability issues.

CIRCUIT DIAGRAM



At the center is the ESP32 microcontroller which serves as the brain of the system, receiving power from two 12V DC adapters through a voltage regulator module (blue component on the left) that steps down voltage to 5V for sensitive electronics. The ESP32 connects to multiple sensors including the DHT11 temperature and humidity sensor (blue module at top), pH sensor module (left side), and light intensity sensor (brown module) through appropriate power, ground, and signal lines shown in red, black, and pink respectively. The microcontroller's GPIO pins connect via control lines to a 4-channel relay module (blue component at center-right) which acts as an interface between low-voltage control signals and high-power devices. Through the relay contacts, the system controls several actuators including a water pump (bottom right), two cooling fans (black squares at bottom), and LED grow lights (yellow tube at bottom left), all powered by the 12V supply when their respective relays are activated by the ESP32.

BLOCK DIAGRAM



The above diagram illustrates an **IoT-based Environmental Control System** where a 12V 2A power supply feeds a buck converter that regulates voltage for an ESP32 microcontroller, which serves as the system's brain with WiFi/Bluetooth connectivity. The ESP32 reads real-time data from a DHT sensor (temperature & humidity) and pH sensor (water quality), then intelligently controls four output devices—two cooling fans, a water pump, and a UV light—through a 4-channel relay module that acts as an electronic switch. A secondary 12V 2A power supply directly powers the high-current actuators (fans, pump, UV light) while the relay provides electrical isolation and switching control based on commands from the ESP32. This automated system is ideal for applications like hydroponics, greenhouse automation, or aquaponics, where it maintains optimal growing conditions by automatically adjusting cooling, irrigation, and lighting based on sensor readings and can be monitored/controlled remotely via internet connectivity.

ADVANTAGES

This automated environmental control system offers numerous significant advantages over traditional manual systems.

First, it provides **24/7 autonomous operation** without human intervention, continuously monitoring environmental conditions and making real-time adjustments to maintain optimal growing parameters, which eliminates the need for constant manual checking and reduces labor costs.

Second, the **IoT connectivity** through ESP32's WiFi/Bluetooth enables remote monitoring and control from anywhere in the world via smartphone or computer, allowing users to check sensor data, receive alerts, and adjust settings even when away from the installation site.

Third, the system ensures **precise environmental control** by responding immediately to sensor readings, maintaining consistent temperature, humidity, and water quality within narrow optimal ranges that would be difficult to achieve manually, leading to better plant growth, higher yields, and reduced crop failure.

Fourth, it offers **energy efficiency** by operating devices only when needed based on actual sensor data rather than fixed schedules, reducing electricity consumption and operating costs.

Fifth, the **scalable and modular design** allows easy expansion by adding more sensors or actuators to the ESP32's available GPIO pins without major system redesign.

Sixth, the **data logging capability** enables long-term tracking of environmental patterns, helping users optimize their growing strategies and identify issues before they become critical.

Seventh, the use of **standardized, affordable components** like ESP32, DHT sensors, and relay modules makes the system cost-effective to build and maintain compared to expensive commercial automation solutions, while the

dual power supply design with buck converter ensures stable, reliable operation and protection of sensitive electronic components.

Additionally, the system provides **predictive maintenance capabilities** by monitoring device performance and power consumption patterns, allowing early detection of failing components like clogged pumps or worn-out fans before complete failure occurs.

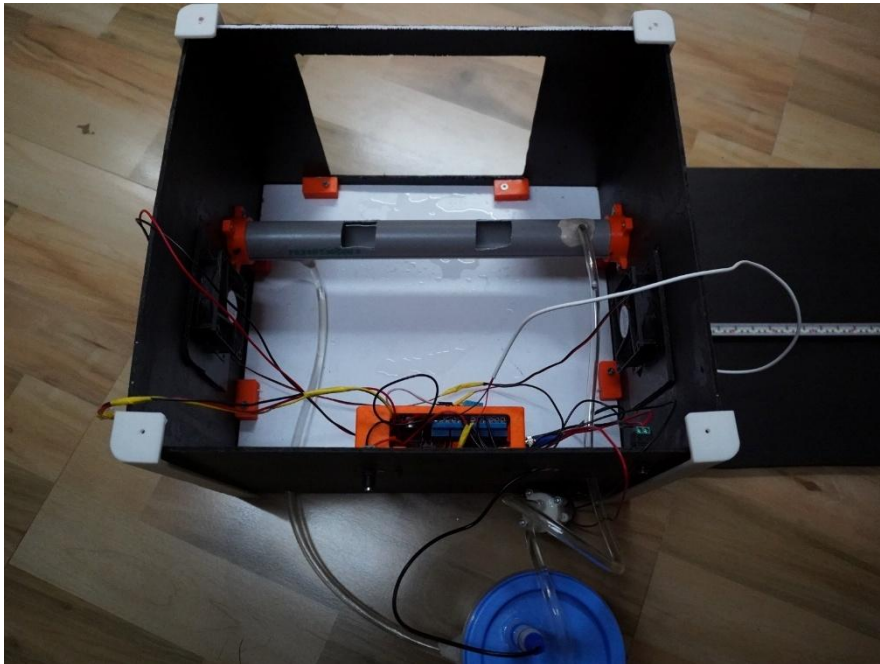
The relay isolation protects the sensitive ESP32 microcontroller from high-voltage spikes and electrical noise generated by motors and pumps, significantly improving system reliability and longevity. **The dual power supply architecture** ensures that critical control circuits remain operational even if one power source fails, providing redundancy and preventing catastrophic system failures.

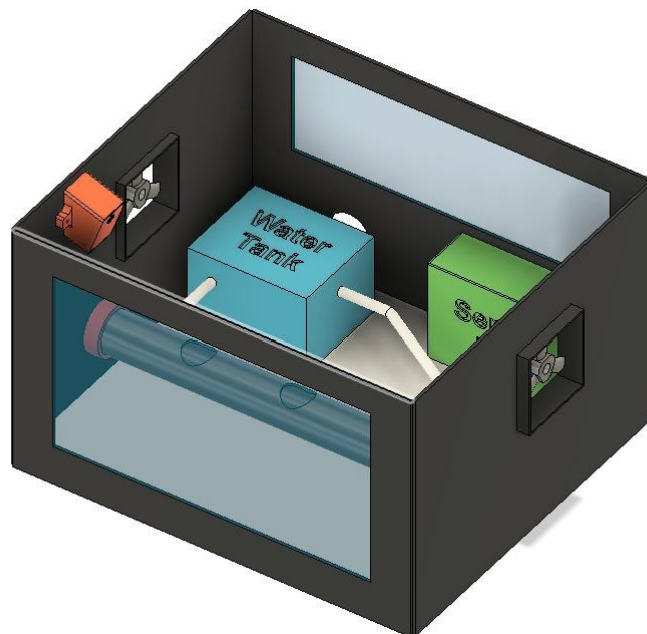
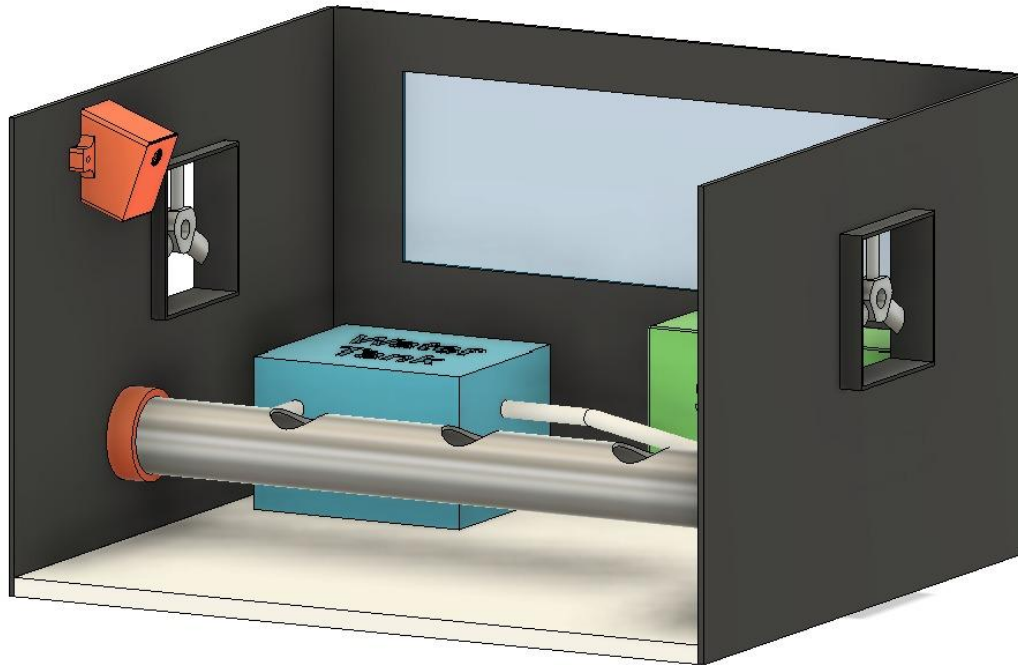
The real-time pH monitoring prevents nutrient imbalances that could harm plants or aquatic life, automatically triggering corrections before damage occurs, which is especially critical in hydroponic and aquaponic systems where water chemistry changes rapidly.

The automated UV light control optimizes sterilization cycles and light exposure for plant growth while minimizing energy waste and extending bulb lifespan. **The intelligent cooling system** with dual fans provides redundancy and allows for graduated temperature control, running one fan for minor adjustments and both for rapid cooling, which is more efficient than single-fan on/off cycling.

Furthermore, the system offers **integration potential** with other smart home or agricultural automation platforms through standard IoT protocols like MQTT, allowing coordination with weather stations, irrigation controllers, or greenhouse management systems for holistic farm automation. **The open-source nature** of ESP32 programming enables complete customization of control algorithms, threshold settings, and response behaviors to match specific crops, environmental conditions, or growing methodologies without vendor lock-in.

RESULTS AND IMAGES





Smart Hydroponic System using IOT

**Smart Agriculture System**
Real-time Monitoring & Control

Websocket Connected Reconnect

Historical Data

Time	pH	Temp (°C)	Humidity (%)
2:45:11 PM	6.27	28.0	67.0
2:45:10 PM	6.35	28.0	67.0
2:45:08 PM	6.24	28.0	67.0
2:45:07 PM	6.24	28.0	67.0
2:45:06 PM	6.30	28.0	67.0
2:45:05 PM	6.24	28.0	67.0
2:45:04 PM	6.25	28.0	67.0
2:45:02 PM	6.31	28.0	67.0
2:45:01 PM	6.36	28.0	66.0
2:45:00 PM	6.27	28.0	66.0
2:44:59 PM	6.25	28.0	66.0
2:44:57 PM	6.27	28.0	66.0
2:44:56 PM	6.19	28.0	66.0

Capture Image

Capture & Analyze



Image loaded (2291 chars)

Status: Ready

Mode: Capture Only

Resolution: 640x480

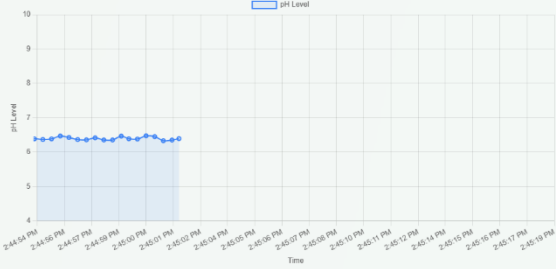
Websocket: Connected

AI Analysis

The plant appears severely stressed, likely dead. The branches are bare and dry, indicating a lack of hydration and potentially a nutrient deficiency. It might be dormancy as well.

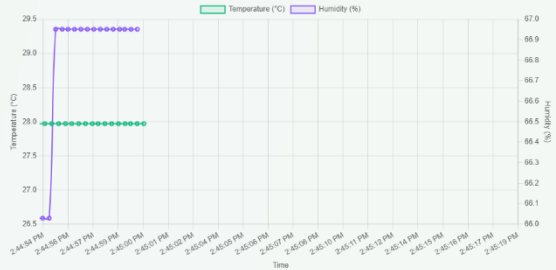
Overall health rat

pH Level Graph



Time	pH Level
2:44:54 PM	6.27
2:44:55 PM	6.35
2:44:56 PM	6.24
2:44:57 PM	6.24
2:44:58 PM	6.30
2:44:59 PM	6.24
2:45:00 PM	6.25
2:45:01 PM	6.31
2:45:02 PM	6.36
2:45:03 PM	6.27
2:45:04 PM	6.25
2:45:05 PM	6.27
2:45:06 PM	6.19

Temp Humidity Graph



Time	Temperature (°C)	Humidity (%)
2:44:54 PM	28.0	67.0
2:44:55 PM	28.0	67.0
2:44:56 PM	28.0	67.0
2:44:57 PM	28.0	67.0
2:44:58 PM	28.0	67.0
2:44:59 PM	28.0	67.0
2:45:00 PM	28.0	66.0
2:45:01 PM	28.0	66.0
2:45:02 PM	28.0	66.0
2:45:03 PM	28.0	66.0
2:45:04 PM	28.0	66.0
2:45:05 PM	28.0	66.0
2:45:06 PM	28.0	66.0
2:45:07 PM	28.0	66.0
2:45:08 PM	28.0	66.0
2:45:09 PM	28.0	66.0
2:45:10 PM	28.0	66.0
2:45:11 PM	28.0	66.0
2:45:12 PM	28.0	66.0
2:45:13 PM	28.0	66.0
2:45:14 PM	28.0	66.0
2:45:15 PM	28.0	66.0
2:45:16 PM	28.0	66.0
2:45:17 PM	28.0	66.0
2:45:18 PM	28.0	66.0
2:45:19 PM	28.0	66.0

Sensor Data

Soil pH

6.28 pH

●

Light Intensity

2.9 %

●

Temperature

28.0 °C

●

Humidity

67.0 %

●

Device Controls

Grow Lights

OFF

Water Pump

OFF

Fan

OFF

APPLICATIONS

This versatile automated environmental control system has numerous practical applications across agriculture, research, and specialized growing environments. **In hydroponics operations**, the system maintains optimal nutrient solution pH levels, controls water circulation pumps, regulates growing chamber temperature and humidity, and manages lighting schedules for maximum plant growth without soil, making it ideal for urban farming, vertical farms, and year-round vegetable production in controlled environments.

For aquaponics systems, it monitors water quality parameters critical for both fish health and plant growth, controls aeration pumps and filters, maintains proper temperature ranges for aquatic species, and ensures the symbiotic ecosystem remains balanced by automatically adjusting conditions when fish waste levels or pH drift from optimal ranges.

In commercial greenhouses, the system automates climate control by managing ventilation fans, misting systems, supplemental lighting, and heating/cooling equipment based on real-time temperature and humidity readings, reducing labor costs while maximizing crop yield and quality for flowers, vegetables, and ornamental plants.

For mushroom cultivation, where precise environmental control is critical, the system maintains the specific humidity levels (often 85-95%), temperature ranges, and fresh air exchange rates required for different mushroom species and growth stages, while UV sterilization prevents contamination that can destroy entire crops.

In research laboratories and university agricultural programs, the system provides consistent, reproducible environmental conditions for plant science experiments, allows researchers to test how specific environmental variables affect growth, enables automated data collection for analysis, and can run multiple test chambers with different parameters simultaneously.

For home gardening enthusiasts and small-scale urban farmers, it transforms basements, spare rooms, or balconies into productive growing spaces with automated climate control, making year-round fresh produce accessible even in apartments or regions with harsh climates.

In nursery operations and seedling propagation, the system maintains the warm, humid conditions needed for germination and early growth stages, automatically adjusts parameters as plants mature, and ensures consistent quality across thousands of seedlings for commercial plant production.

For exotic plant collections and orchid growing, where different species require very specific temperature and humidity ranges, the system can maintain the precise

microclimates needed for rare or difficult species that would otherwise fail in standard growing conditions.

In cannabis cultivation facilities, both medical and recreational, the system provides the precise environmental control required for legal compliance, optimal cannabinoid production, and consistent crop quality while maintaining detailed data logs for regulatory requirements.

For indoor algae farming and spirulina production, it controls water circulation, maintains optimal pH for photosynthesis, manages lighting cycles, and regulates temperature for maximum biomass production in this emerging sustainable protein source industry.

In vertical farming operations, where multiple growing levels require independent climate zones, the system can control separate environmental parameters for different crops or growth stages within the same facility, maximizing space efficiency and production diversity.

For seed germination chambers, it provides the precise temperature and humidity control needed for high germination rates in commercial seed production or seed bank preservation.

In controlled environment agriculture (CEA) facilities, it integrates with larger building management systems to optimize resource usage across entire growing operations. **For educational institutions**, it serves as a hands-on learning tool for students studying agricultural technology, IoT systems, environmental science, and sustainable farming practices.

In pharmaceutical plant cultivation, where specific environmental conditions affect active compound production in medicinal plants, the system ensures consistent pharmaceutical-grade quality. **For insect farming operations** producing protein for animal feed or human consumption, it maintains the specific temperature and humidity requirements for crickets, mealworms, or black soldier flies. **In tissue culture laboratories**, it provides the sterile, controlled environment necessary for plant cloning and micropropagation.

Finally, in disaster relief and food security applications, these portable automated systems can be deployed to establish rapid food production in refugee camps, remote communities, or areas affected by climate change where traditional agriculture has become unreliable, providing fresh produce with minimal water usage and no dependence on arable land.

CONCLUSION

This IoT-based environmental control system represents a significant advancement in modern agriculture and controlled environment management, demonstrating how automation and connectivity can revolutionize traditional growing practices. By integrating an ESP32 microcontroller with DHT temperature-humidity sensors, pH sensors, and a 4-channel relay module, the system creates an intelligent framework that continuously monitors critical environmental parameters and automatically controls cooling fans, water pumps, and UV lighting to maintain optimal growing conditions. The thoughtful architecture featuring dual power supplies, buck converter voltage regulation, and electrical isolation ensures reliable, safe operation while protecting sensitive components from voltage spikes and electrical interference.

The system's versatility spans diverse applications including hydroponics, aquaponics, commercial greenhouses, mushroom cultivation, vertical farming, research laboratories, and home gardening. Each application benefits from precise environmental control that would be difficult or impossible to achieve manually. The economic advantages are substantial: energy savings through on-demand operation, reduced labor costs, optimized water usage, and improved crop yields with lower failure rates. The use of affordable, standardized components makes this technology accessible to small-scale growers and developing regions, democratizing advanced agricultural techniques previously available only to large commercial operations.

Beyond immediate practical benefits, this system addresses critical global challenges including food security, climate change adaptation, and sustainable resource management. It enables year-round food production independent of weather conditions, reduces water consumption through precise irrigation, minimizes chemical usage through better environmental management, and allows agriculture in non-traditional spaces like urban environments and regions where conventional farming is impractical.

As climate change increasingly disrupts traditional agricultural patterns and global population continues growing, intelligent automated systems like this transition from innovative solutions to essential infrastructure for sustainable food production.

In conclusion, this environmental control system exemplifies the convergence of agriculture and technology, proving that smart automation improves efficiency, productivity, and sustainability while remaining practical and affordable. The modular, expandable design allows customization for specific needs, while open-source ESP32 programming prevents vendor lock-in and encourages innovation. Whether deployed in a basement grow room, commercial greenhouse, or research facility, this system delivers consistent, reliable environmental control that transforms growing from an art into a precise science. As we face mounting pressure to feed a growing population with fewer resources, such intelligent systems represent not just an improvement but a fundamental reimagining of sustainable agriculture, accessible to anyone with basic electronics knowledge and commitment to efficient growing practices.